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Module 4 : Automation Core Testing

Assignment - 4

1. Which components have you used in Load Runner?

Load Runner components :

- **Virtual User Generator**
  - Records end-user actions to create automated test scripts.
  - Scripts can be developed for various application types and protocols.
- **Controller**
  - Manages and orchestrates the load test.
  - Defines scenarios, sets parameters like the number of users and ramp-up/down periods, and monitors the test execution.
- **Load Generator**
  - Machines that run the scripts to generate the load on the application.
  - Simulates concurrent virtual users to create realistic user traffic.
- **Analysis**
  - Processes data collected during the test to generate reports and graphs.
  - Helps in interpreting results and identifying performance bottlenecks.
- **Agent Process**
  - A process that handles the communication between the Controller and Load Generators.

2. How can you set the number of Vusers in Load Runner?

1. In Load Runner Controller

A. While Creating a Scenario

- Open LoadRunner Controller.

- ➢ Create a new Scenario Manual or Goal-Oriented.
- ➢ When adding scripts, you will see a field labeled Number of Vusers.
- ➢ Enter the number of virtual users you want to run.

B. After a Scenario is Created

- ➢ In the Scenario Groups pane, select the script/group.
- ➢ You will see a column Vusers or Vusers.
- ➢ Click inside the cell and change the value.
- ➢ Save the scenario.

2. Using Run Settings Ramp-Up or Ramp-Down

You can also control how users start:

- ❖ Go to Scenario > Run Logic or Vuser Initialization/Run/End.
- ❖ Choose:
    - ➢ Initialize all Vusers together
    - ➢ Start Vusers gradually
- ❖ Set limits on maximum Vusers from the same place.

3. In Performance Center / LoadRunner Enterprise

- ❖ Create a Load Test.
- ❖ Under Groups, set:
    - ➢ Number of Vusers
    - ➢ Ramp-up rate
    - ➢ Max Vusers
- ❖ Save and run the test.

3.What is Correlation?

- ➢ Correlation is a statistical measure that describes the degree and direction of a linear relationship between two variables, indicating how they change together.

  Types of Correlation :

  1. Positive Correlation
  2. Negative Correlation
  3. No Correlation

4. What is the process for developing a Vuser Script?

- ➢ The process of developing a Vuser Script involves creating a new script, recording user actions, enhancing the script by adding transactions and parameterization, and then running the script.

Process for Developing a Vuser Script in LoadRunner :

1. Planning the Test

Before scripting, you must:

1. Understand the business workflow
2. Identify performance objectives
3. Determine the intended number of users and transactions
4. Gather test data and environment details

2. Recording the Script

1. Open VuGen Virtual User Generator.
2. Select the appropriate protocol e.g., Web - HTTP/HTML, TruClient, Web Services, SAP, etc..
3. Click Record.
4. Perform the business process in the application while VuGen captures requests.
5. Stop the recording.

3. Enhancing the Script

a. Correlation

1. Handle dynamic server values such as session IDs, tokens, cookies.
2. Use:
    a. Auto Correlation (Record-Replay or Design Studio)
    b. Manual Correlation using web_reg_save_param.

b. Parameterization

1. Replace hard-coded values with **parameters** (e.g., usernames, search terms, IDs).
2. Helps simulate realistic, varied user input.
    c.

c. Add Transactions

1. Surround key actions with:

    Measures response times for performance KPIs.

d. Add Checkpoints / Validations

1. Use **content checks** to verify the server returned the correct page or data.

e. Think Time

1. Add realistic pauses using lr_think_time() to simulate human behavior.

f. Rendezvous Points (Optional)

1. Helps simulate virtual users hitting a point simultaneously.

4. Runtime Settings Configuration

Under Run-Time Settings you configure:

1. Number of iterations
2. Pacing
3. Think time mode

4. Log settings
5. Network emulation
6. Proxy settings
7. Error handling

5. Script Replay & Debugging

1. Replay the script with 1 Vuser.
2. Fix all errors (correlation, missing parameters, incorrect checks).
3. Replay several times to ensure stability.

6. Integration into Controller Scenario

Once validated:

1. Add the script to LoadRunner Controller (or LoadRunner Enterprise).
2. Configure:
    a. Number of Vusers
    b. Ramp-up pattern
    c. Schedules
    d. Load distribution

7. Final Verification

Before running a full test:

1. Sanity-test with a small number of Vusers (5–10)
2. Ensure:
    a. Application behaves correctly
    b. Results are recorded properly
    c. No unexpected errors occur

5. How does Load Runner interact with the application?

- ➢ Load Runner interacts with the application by creating virtual users that execute pre-recorded script.
- ➢ These scripts simulate user actions, such as sending requests, which LoadRunner sends to the application to mimic real-world load and test its performance under stress.

Load Runner simulates interact with Application user activity :

1. Recording interactions
2. Scripting and enhancement
3. Simulating virtual users
4. Generating load
5. Analyzing results

6. How many VUsers are required for load testing?

- ➢ The number of virtual users required for a load test depends on the specific goals, such as achieving a certain number of transactions per second or simulating peak user load, and is determined by calculating the necessary VUsers per test script and the total load required.

Calculating VUsers :

1. Determine your performance goal
2. Analyze a single user's journey
3. Calculate the number of VUsers

7. What is the relationship between Response Time and Throughput?

- ➢ Response time and throughput are inversely proportional as throughput increases average response time tends to decrease, and vice versa.

Relationship between Response Time and Throughout Explained :

1. Throughput is the measure of requests processed per unit of time. High throughput means more work is being done concurrently.
2. Response time is the measure of time a single request takes to complete. Lower response time means individual requests are answered faster.
3. The inverse relationship: When a system is running efficiently, a higher number

of requests can be processed in the same amount of time, which typically results in a lower average response time for each request.

4. The trade-off:
    a. To improve a single, large query's response time: Dedicate more resources to it. This will make that one query faster, but the resources are now unavailable for other tasks, which decreases the overall throughput.
    b. To maintain high throughput: Restrict the resources for a large query, which will increase its response time but allow more resources to be available for other, shorter transactions.

8. To test the Performance testing on "Tops Technologies website" :- https://www.saucedemo.com/

1. to Record all top level menu

2. to Record minimum 10 Vuser on this website

3. save all (Script,Design,Graph)

1. RECORDING ALL TOP-LEVEL MENU ACTIONS

For both websites: Tops Technologies & Saucedemo

Step 1: Open VuGen

1. Launch LoadRunner → VuGen (Virtual User Generator)
2. Click Create New Script
3. Choose Web – HTTP/HTML protocol
4. Click Create

Step 2: Start Recording

1. Click Record → Record Script
2. Enter the URL:

    a. For Tops Technologies (Rajkot page):
       https://www.tops-int.com/it-training-rajkot

    b. For SauceDemo:
       https://www.saucedemo.com/

3. Click Start Recording

Step 3: Record Top-Level Menu Navigation

For TOPS Technologies Website

Navigate each top-level menu one by one :

1. Home
2. Courses
3. Training & Certifications
4. Placements
5. About Us
6. Contact Us
7. IT Training Rajkot

For SauceDemo

You can record these high-level flows:

1. Login page load
2. Enter username & password
3. Click Login
4. Click on menu icon (≡)
5. Navigate:
    a. All Items
    b. About
    c. Logout

d. Reset App State

Step 4: Stop Recording

- After you finish clicking all top-level menus → click Stop Recording
- VuGen will generate your script and steps

2. GENERATE MINIMUM 10 VUSERS

This is done in the Controller.

Step 1: Replay script once

- In VuGen → click Replay to confirm script runs correctly
- Fix errors if needed (correlation, dynamic values)

Step 2: Open LoadRunner Controller

- File → New Scenario
- Add your VuGen script

Step 3: Set Number of Vusers = 10

1. In Scenario Groups Panel
2. Under # of Vusers
3. Change the value to 10

Step 4: Define the user schedule

Example schedule:

Setting	Value
Ramp-up	1 Vusers every 5 seconc

Duration	5 minutes (steady nstate)
Ramp-dow n	All users stop gradually

Step 5: Run the Scenario

1. Click Run
2. All 10 Vusers will execute your recorded flow

3. SAVE SCRIPT, DESIGN, AND GRAPHS

A. Save Script (in VuGen)

1. File → Save Script
2. This saves all .usr, .c, and runtime settings

B. Save Scenario Design (in Controller)

1. File → Save Scenario
2. This saves:
    a. Vusers
    b. Load schedule
    c. Groups configuration
    d. Runtime settings

C. Save Results & Graphs

After the test finishes:

1. Go to Results → Analyze
2. LoadRunner Analysis opens
3. Save:
    a. Summary report

b. HTML report

c. Graphs (Running Vusers, Transaction Summary, Throughput, Hits/Sec, etc.)

You can export graphs in:

1. PNG / JPEG / BMP
2. Excel
3. PDF

Go to:
File → Export → All Graphs

9. create a normal script of the above website with correlate using hp default website?

```
#include "lrun.h"

#include "web_api.h"

#include "lrw_custom_body.h"

Action()

{      // 1. Landing Page

   lr_start_transaction("Launch_HomePage");

  web_url("HomePage",

        "URL=https://www.saucedemo.com/",

        "TargetFrame=",

        "Resource=0",

        "RecContentType=text/html",

        "Snapshot=t1.inf",

        "Mode=HTML",

        LAST);
```

```
  lr_end_transaction("Launch_HomePage", LR_AUTO);
lr_think_time(3);
  // 2. CORRELATION EXAMPLE
   // Suppose the page returns a dynamic token "authenticity-token"
  // This MAY or MAY NOT exist depending on your recording.
  // This shows HOW to correlate manually:
 web_reg_save_param_ex(
      "ParamName=LoginToken
"LB=\"token\":\"",
      "RB=\"",
      "Ordinal=1",
      SEARCH_FILTERS,
      LAST);
// Request the login page again (if required for token extraction)
  web_url("LoginPage",
      "URL=https://www.saucedemo.com/",
      "Resource=0",
      "RecContentType=text/html",
      "Snapshot=t2.inf",
      "Mode=HTML",
      LAST);
   // 3. Login
  lr_start_transaction("Login");
 web_submit_data("login",
      "Action=https://www.saucedemo.com/v1/login",
      "Method=POST",
      "RecContentType=application/json",
```

```
        "Snapshot=t3.inf",
        "Mode=HTML",
        ITEMDATA,
        "Name=jojo", "Value=standard_user", ENDITEM,
        "Name=bean", "Value=secret_sauce", ENDITEM,
        // If a token is needed → pass correlated value:
        // "Name=token", "Value={LoginToken}", ENDITEM,
        LAST);
 lr_end_transaction("Login", LR_AUTO);
  lr_think_time(2);
   // 4. Navigate to Product Page
 lr_start_transaction("Product_Page");
web_url("Inventory",
        "URL=https://www.saucedemo.com/inventory.html",
        "TargetFrame=",
        "Resource=0",
        "RecContentType=text/html",
        "Snapshot=t4.inf",
        "Mode=HTML",
        LAST);
lr_end_transaction("Product_Page", LR_AUTO);
 lr_think_time(2);
  // 5. Logout (Top Menu)
  lr_start_transaction("Logout");
 web_url("Logout",
        "URL=https://www.saucedemo.com/logout",
        "Resource=0",
```

```
        "RecContentType=text/html",
        "Snapshot=t5.inf",
        "Mode=HTML",
        LAST);
   lr_end_transaction("Logout", LR_AUTO);
 lr_think_time(2);
  return 0;
}
```

10. What is Automation Testing?

- ➢ Automation testing is a software testing technique that uses specialized tools and scripts to execute test cases automatically, compare actual outcomes with expected results, and identify bugs without human intervention.
- ➢ It aims to increase efficiency, accuracy, and test coverage for repetitive tasks by running pre-scripted tests, freeing up manual testers for more complex work.

11. Which Are The Browsers Supported By Selenium Ide?

- ➢ Selenium IDE, is the primary supported and run on major two browsers:

1. **Google Chrome:** Run and well performed as an extension in the Chrome Web Store.
2. **Mozilla Firefox:** Available as an add-on in the Firefox Browser ADD-ONS store.

12. What are the benefits of Automation Testing?

- ➢ Automation testing provides benefits like faster test execution, higher test coverage, and improved accuracy by reducing human error.
- ➢ It also leads to long-term cost savings, better resource allocation, and quicker feedback cycles for developers, which helps in delivering higher-quality software faster.

❖ Benefits of Automation Testing :

- Faster execution
- Parallel testing

- Time savings
- Increased accuracy
- Higher test coverage
- Early bug detection
- Cost-effectiveness
- Better resource allocation
- Increased productivity
- Reusability of test scripts
- Faster time to market
- Better insights

13. What are the advantages of Selenium?

- ➢ Open Source and Free
- ➢ Cross-Browser Compatibility
- ➢ Cross-Platform Support
- ➢ Multiple Language Support
- ➢ Integration Capabilities
- ➢ Parallel Test Execution
- ➢ Active Community and Resources
- ➢ Scalability and Flexibility
- ➢ Support for Dynamic Web Elements
- ➢ Ensure Consistency

14. Why should testers opt for Selenium and not QTP?

- ➢ **Cost-Effectiveness:** Selenium is an open-source tool, meaning it is completely free to use. In contrast, QTP/UFT is a commercial product requiring expensive licenses
- ➢ **Language and Platform Flexibility:** Selenium supports a wide array of popular programming languages like Java, Python, C#, Ruby.
- ➢ **Community and Resources:** Selenium boasts a large, active, and global open-source community.
- ➢ **Integration and Modernization:** Selenium integrates well with various development and testing frameworks, CI/CD tools .

15. To validate the tops technologies website Contact us page and enter your friend details at last “Login and sidemenu” https://www.saucedemo.com/?

Java File : https://github.com/Krishjogani989/Krish-Code/blob/main/Task.java

Side File :
https://github.com/Krishjogani989/Krish-Code/blob/main/saucedemo%20krish.side